UNIT 5
BIOINFORMATICS

Connect to the topic

1. Look at the photo. What kind of research is the woman doing?
2. How do computers help in analyzing genetically modified food??
3. What is bioinformatics, and why is it important?
4. What kind of data do researchers collect about food?
5. How can AI help in food research?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about the biotechnology and discuss the questions below.

1. What is biotechnology, and how does it relate to bioinformatics?
2. What are the eight categories of biotechnology mentioned in the video?
3. What is the role of bioinformatics in analyzing DNA sequencing?
4. Should people be allowed to edit their own genes? Why or why not?
5. Could bioinformatics help prevent future pandemics? How?
6. How can bioinformatics contribute to forensic science and criminal investigations?

UNIT 5.1 EXPLORING BIOINFORMATICS
READING AND VOCABULARY

Task 1. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

1. **DNA sequence (phr)** – the order of letters (A, T, C, G) in a DNA that carries genetic information
2. **computational model (phr)** – a computer program that simulates how something works in real life.
3. **genetic disorder (phr)** – a disease caused by a problem in a person's DNA.
4. **personalized medicine (phr)** – treatment designed to fit a person's unique genes and health.
5. **genetic makeup (phr)** – all the genes that a person or living thing has.
6. **gene mapping (phr)** – the process of finding the location of genes on a DNA.
7. **high-throughput (adj)** – able to do many tests or process lots of data very quickly.
8. **resistance (n)** – the ability to fight against something.
9. **endangered species (phr)** – animals or plants that are at risk of disappearing forever.
10. **microbial community (phr)** – a group of tiny living things (like bacteria) that live together in the same place.

1. Scientists used a ______________ to predict how a new virus would spread in the human body.
2. The research lab used _____________ machines to study thousands of samples in one day.
3. The new drug was developed as part of_____________, based on the patient's unique DNA.
4. By analyzing your_____________, doctors can understand your risk for certain diseases.
5. Bioinformatics helps identify the exact DNA ____________ that may lead to inherited health problems.
6. Experts are using ___________ to understand which genes help animals survive in changing climates.
7. In your gut, there is a complex _____________that helps digest food and keeps you healthy.
8. The medical team discovered that a _____________ was causing the child's rare symptoms.
9. Because of antibiotic overuse, some bacteria have developed _____________ to treatment.
10. Scientists are working to protect _______________ like sea turtles and mountain gorillas through genetic research.

Task 2. READ FOR DETAILS. Read the text and answer the questions below.

What Is Bioinformatics?

Bioinformatics is a new science where computer science, biology, and data analysis come together. It focuses on the examination of genetic data, such as DNA sequences, using complex algorithms and mathematical models. Instead of studying genes through a laboratory microscope, scientists now use computers to cut, compare, and predict hidden messages in our DNA.

One of the primary goals of bioinformatics is to understand how genes work and how genes link to disease. For example, researchers can study with the aid of bioinformatics tools mutations that result in cancer or other genetic diseases. This helps doctors to develop personalized medicine – medicines designed to a patient's unique genetic makeup.

Bioinformatics also helps scientists in gene mapping, protein structure prediction, and the design of databases that hold biological information. With the help of cloud computing and high-throughput platforms, large quantities of information may be processed in a matter of minutes.

Bioinformatics is used in healthcare only. It is an essential usage in agriculture, as biologists study plant DNA to improve crops, increase disease resistance, and supply better food quality. It is also used in environmental science to trace, examine microbial communities, and unravel how ecosystems respond to climate change. Since the world is now confronted with problems of multiple dimensions, bioinformatics gives us powerful problem-solving tools based on the language of life – data.

(adapted from https://www.britannica.com/science/bioinformatics)

1. What three areas of science come together in bioinformatics?
2. Why do scientists use computers in bioinformatics instead of working only in a lab?
3. How can bioinformatics help doctors treat diseases?
4. What is personalized medicine?
5. What are two other tasks bioinformatics can perform besides disease research?
6. How do high-throughput technologies help bioinformatics?
7. In what way is bioinformatics used in agriculture?
8. How does bioinformatics support environmental science?

Task 3. WORK WITH WORDS. Complete the sentence with the correct word from the box.

sequence alignment ▪ data visualization ▪ biological sample ▪ species ▪ pattern recognition ▪ DNA sequencing ▪ evolutionary relationships ▪ genome ▪ computational tool ▪ mutation

1. Scientists use _______ to find changes in an organism's DNA.
2. A _______ contains all the information needed to build and maintain an organism.
3. To understand genetic data better, researchers often use _______.
4. When a _______ happens, it can lead to new traits in a population.
5. A _______ is a group of organisms that are similar and can breed together.
6. _______ help scientists understand how animals and plants are connected through evolution.
7. _______ helps scientists notice similarities or important changes in large data sets.
8. A _______ might include blood, saliva, or tissue.
9. During _______, scientists arrange DNA sequences to find similarities or differences.
10. Scientists often use a _______ to process and analyze large genetic databases.

Task 4. WORK WITH WORDS. Choose the word or phrase that doesn't belong in each group.

Which one doesn't belong?

1. DNA sequencing / genome / data visualization / genetic makeup
2. pattern recognition / mutation / sequence alignment / resistance
3. personalized medicine / biological sample / microbial community / species
4. genetic disorder / computational tool / mutation / species
5. evolutionary relationships / personalized medicine / genetic makeup / microbial community
6. biological sample / DNA sequencing / sequence alignment / genetic disorder
7. resistance / genome / personalized medicine / evolutionary relationships
8. pattern recognition / data visualization /genetic makeup / computational tool

Task 5. COLLABORATE. Work in pairs. Each of you will get a different short text about a real-life use of bioinformatics. Read your text carefully, and then explain it to your partner in your own words. After explaining, ask your partner 2–3 questions to check if they understood.

Use these question starters to help:

- What was the main point about...?
- Can you remember why...?
- How does ... help scientists?

- What is one example of...?

Student A: DNA Sequencing and Ancient Animals
Scientists use DNA sequencing to study ancient animals like mammoths and saber-toothed cats. By analyzing old bones or hair, they can find the genetic code of species that lived thousands of years ago. Bioinformatics helps by comparing this ancient DNA to the DNA of modern animals, helping scientists understand evolutionary relationships and why some species became extinct. Thanks to computational tools, these comparisons are faster and more accurate.

Student B: Personalized Medicine and Genetic Makeup
Personalized medicine uses information about a person's genetic makeup to choose the best treatment for them. For example, some people have genetic differences that make certain drugs less effective. Bioinformatics helps doctors by analyzing huge amounts of genetic data and finding patterns that show which medicine will work best. This can make treatments faster, safer, and more successful.

LISTENING

Task 6. LISTEN FOR DETAILS. Scan the QR code and listen to the dialogue between Professor Flint and Max. Then answer the questions below.

1. What is the basic definition of bioinformatics that Professor Flint gives?
2. Besides genes, what other things can bioinformatics be used to study?
3. How does bioinformatics help protect endangered animals?
4. What example does Professor Flint give about tracking health and population?
5. Can beginners help in bioinformatics research? How?
6. What is the first task Professor Flint says Max will do?

Task 7. COMMUNICATE. Work in pairs and role play the following situation. Max meets his friend Alex the day after joining the research project. Alex is curious about what bioinformatics is and what Max will be doing in the lab.
Student A: You are Max. Explain the research project to your friend using information from the audio.
Student B: You are Alex. Ask questions to learn more.

WATCHING

Task 8. EXLORE THE WORDS. Read the sentences with bold words from the video you are going to watch. Then complete each definition with the correct form of one of the words.

1. Bioinformatics can help scientists **sort through data** about everything from DNA to weather conditions.
2. You might collect information from a **public data set** available online.
3. Many scientists get help from **community science projects** where ordinary people collect important information.
4. Algorithms require you to **communicate instructions to a computer** in a clear way.
5. You've probably **run into** algorithms while browsing on TikTok or YouTube.
6. Your **viewing habits** determine which videos appear on your homepage.
7. Algorithms can **be fed all kinds of instructions** depending on the problem they need to solve.

8. Bioinformatics helps scientists **acquire the data** they need for analysis.
9. Genetic sequences are captured in **overlapping chunks** because they're too long to read all at once.
10. Computers help **assemble a jigsaw puzzle** out of small DNA pieces to form the full sequence.
11. Completing the human genome was a massive **scientific undertaking**.
12. Researchers wanted to understand what **make humans tick** by studying DNA.

Now complete each definition with the correct form of one of the words.

a) __________ discover what causes people to act in certain ways
b) __________ collect or obtain information for a study or project
c) __________ examine large amounts of information carefully to find what you need
d) __________ research projects where the general public helps gather data
e) __________ put together many small pieces to make a complete whole
f) __________ send a set of steps or commands to a computer so it can solve a problem
g) __________ encounter something, usually unexpectedly
h) __________ a collection of information that is open for everyone to use
i) __________ patterns or preferences shown by the videos or shows a person watches
j) __________ small sections that partly cover the same information and fit together
k) __________ be given many different types of directions or rules
l) __________ a major research project that requires a lot of time and effort

Task 9. WATCH FOR DETAILS. Scan the QR code and watch the first part of the video "How Data Saves Lives". Decide whether the following statements are True or False.

1. Dr. Sammy believes that taking a pizza survey will immediately give clear results.	True / False
2. Bioinformatics can only be used to study living organisms, not weather conditions.	True / False
3. Scientists often rely on community science projects to supplement their data.	True / False
4. According to the video, all bioinformatics data sets must be collected personally by scientists.	True / False
5. Algorithms can only be created and used by professional programmers.	True / False
6. The YouTube algorithm showed this Crash Course video randomly to viewers.	True / False
7. Computers help scientists by organizing overlapping DNA sequences into a correct order.	True / False
8. The data about jellyfish in a home aquarium was collected during the filming of the video.	True / False
9. In 2022, researchers managed to decode the entire human DNA sequence in a few months.	True / False
10. Even after decoding human DNA, scientists still struggled to understand its full meaning.	True / False

Task 10. WATCH FOR DETAILS. Scan the QR code and watch the second part of the video "How Data Saves Lives". Choose the correct answers for questions.

1. Why is cracking the DNA code not easy to do by hand?
   A. It is encrypted with special symbols.
   B. DNA letters change every day.
   C. DNA has about 3 billion letters.
   D. The DNA sequence is invisible to humans.
2. What is the main role of bioinformatics in studying DNA sequences?
   A. To print DNA on paper.
   B. To compare sequences and find patterns.
   C. To design new species.
   D. To delete harmful genes.
3. What does the BRCA1 gene normally help prevent?
   A. Heart attacks
   B. Hair loss
   C. Allergies
   D. Tumor growth
4. What can happen if a single letter in the BRCA1 gene is wrong?
   A. Increased risk of cancer
   B. Immediate death
   C. Faster cell repair
   D. Better muscle growth
5. What did scientists do to better understand BRCA1 mutations?
   A. Asked patients about their health.
   B. Removed the gene from humans.
   C. Developed a new type of medicine.
   D. Compared thousands of gene variants.
6. What is a transcriptome?
   A. A collection of all RNA molecules at a given time.
   B. A complete DNA sequence.
   C. A type of mutation.
   D. A tool for deleting bad genes.
7. Why can't humans manually analyze transcriptomes?
   A. Because transcriptomes are invisible.
   B. Because transcriptomes are too complex.
   C. Because humans don't have enough RNA.
   D. Because computers are faster at printing.
8. How can bioinformatics help with agriculture?
   A. By speeding up crop harvesting.
   B. By identifying traits like disease resistance.
   C. By inventing new fruits.
   D. By predicting weather changes.
9. What technology helped researchers analyze Zika virus samples in the field?
   A. Giant supercomputers.
   B. Handheld nanopore technologies.
   C. Traditional microscopes.
   D. Satellite imaging.
10. What did the DNA analysis of Zika virus reveal?
   A. The virus was not dangerous.

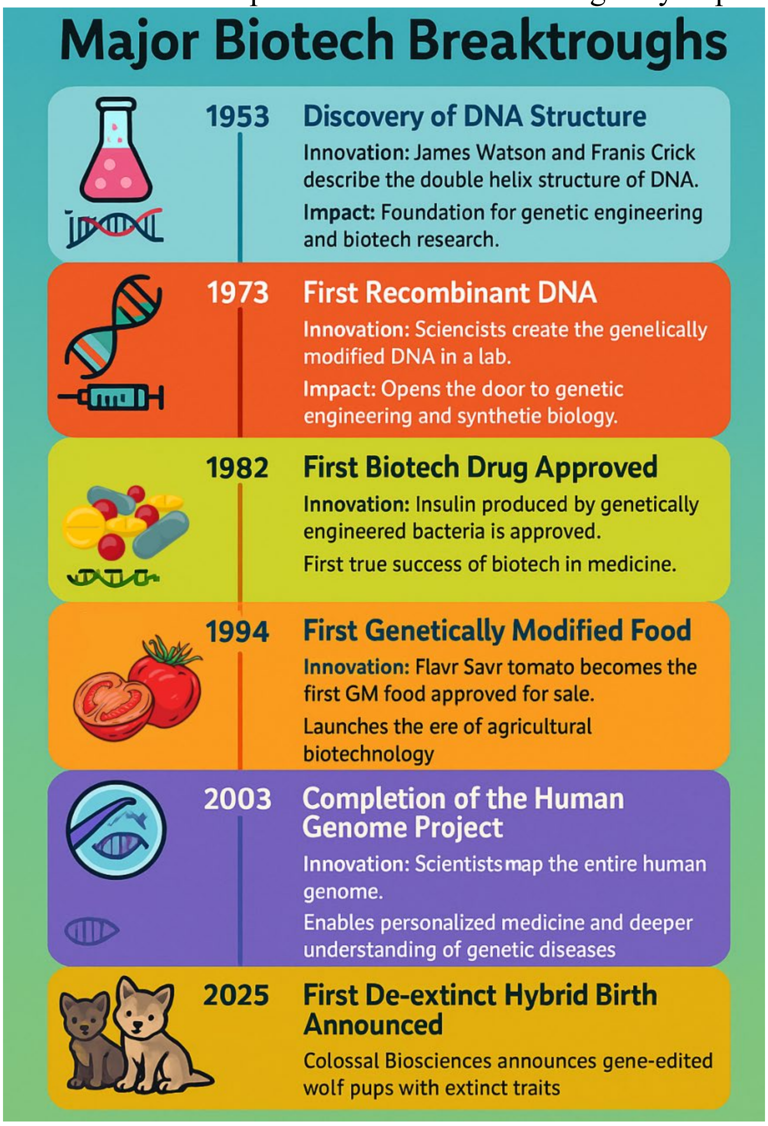
B. The virus had just arrived in Minas Gerais.
C. The virus had been circulating for over a year before detection.
D. The virus was only spread by water.

Task 11. WORK WITH WORDS. Watch the second part again and match the words to make a collocation from the video.

1. **suppress**	a) techniques
2. **compile**	b) tumors
3. **treatment**	c) the device
4. **expressed**	d) plan
5. **RNA**	e) genes
6. **disease**	f) molecules
7. **genetic modification**	g) journey and timeline
8. **life-saving**	h) information
9. **feed**	i) all data
10. **outbreak's**	j) resistance

SPEAKING

Task 12. COMMUNICATE. Work in pairs. Rank the breakthroughs by importance. Explain why.



Task 13. COMMUNICATE. Work in pairs and role play the following situation.

Anna has just completed an internship at a bioinformatics company.
Her friend Mia wants to know what tasks and challenges she faced.

Student A: You are Anna. Describe your internship experience, the tasks you completed, and any problems you helped solve. Mention examples like analyzing genetic data, creating software tools, working with large datasets, helping in drug discovery, facing issues like data errors or software bugs.

Student B: You are Mia. Ask questions about Anna's work and how bioinformatics was used. Ask about the types of projects Anna worked on, what tools she used, what skills were important, and what she found difficult or interesting.

Task 14. COLLABORATE. Work in pairs. Read each case study carefully. Identify the main problem and suggest possible bioinformatics solutions. Think about what tools, methods, or strategies could be used. Be ready to explain your ideas to the class or discuss them in small groups.

CASE STUDY 1

A small town is facing an outbreak of a rare disease. The symptoms are severe, but traditional lab tests can't identify the cause. Doctors suspect a new virus might be involved.

- What bioinformatics tools or methods could help identify the virus?
- How would you organize and analyze the data?

Suggest possible next steps after identification.

CASE STUDY 2

A conservation team has collected DNA samples from a nearly extinct animal species. They hope to understand the genetic diversity in the remaining population to plan breeding programs, but the data is messy and incomplete.

- How could bioinformatics help clean, organize, and analyze this genetic data?
- What strategies would you recommend to improve genetic diversity in the breeding plan?

CASE STUDY 3

A hospital reports that many patients are infected with bacteria that don't respond to common antibiotics. Researchers have sequenced the genomes of the bacteria but are overwhelmed by the amount of genetic data.

- How can bioinformatics help find the genes responsible for antibiotic resistance?
- What bioinformatics tools or databases would you use?

Suggest one way to use this information to improve treatment.

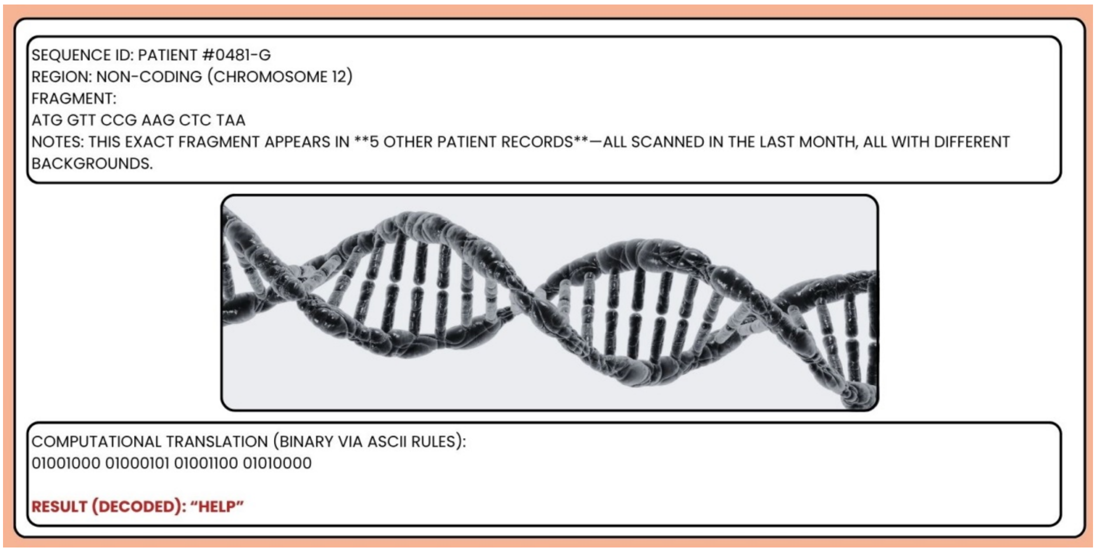
Task 15. COLLABORATE. Take 1 role card, walk around the class, asking and answering questions to complete the grid below.

Name	Field	Tools Used	Data Type	Goal / Application
Dr. Miller				
Dr. Rivas				
Dr. Blake				
Dr. Kwon				
Dr. Tanaka				

CARD 1 Geneticist (Dr. Miller) You study: Human DNA You use: DNA sequencers You analyze: ? You work on: ?	**CARD 2** Geneticist (Dr. Miller) You study: Human DNA You use: ? You analyze: Genetic variants You work on: ?	**CARD 3** Geneticist (Dr. Miller) You study: ? You use: DNA sequencers You analyze: Genetic variants You work on: Identifying inherited disorders
CARD 4 Epidemiologist (Dr. Rivas) You study: ? You use: Genome tracking software You analyze: Virus mutations in outbreaks You work on: ?	CARD 5 Epidemiologist (Dr. Rivas) You study: Virus spread in populations You use: ? You analyze: ? You work on: Controlling epidemics	CARD 6 Epidemiologist (Dr. Rivas) You study: Virus spread in populations You use: Genome tracking software You analyze: Virus mutations in outbreaks You work on: ?
CARD 7 Forensic Scientist (Dr. Blake) You study: Genetic evidence You use: ? You analyze: Genetic samples from crime scenes You work on: ?	CARD 8 Forensic Scientist (Dr. Blake) You study: ? You use: DNA matching algorithms You analyze: ? You work on: Solving crimes	CARD 9 Forensic Scientist (Dr. Blake) You study: Genetic evidence You use: DNA matching algorithms You analyze: Genetic samples from crime scenes You work on: ?
CARD 10 Microbiome Analyst (Dr. Kwon) You study: ? You use: Metagenomic sequencing tools You analyze: DNA of bacteria You work on: ?	CARD 11 Microbiome Analyst (Dr. Kwon) You study: Gut bacteria You use: ? You analyze: ? You work on: Improving digestive health	CARD 12 Microbiome Analyst (Dr. Kwon) You study: Gut bacteria You use: Metagenomic sequencing tools You analyze: DNA of bacteria You work on: ?
CARD 13 Cancer Researcher (Dr. Tanaka) You study: Cancer genes You use: ? You analyze: Genetic changes in tumors You work on: ?	CARD 14 Cancer Researcher (Dr. Tanaka) You study: ? You use: Mutation databases You analyze: ? You work on: Developing personalized treatments	CARD 15 Cancer Researcher (Dr. Tanaka) You study: Cancer genes You use: Mutation databases You analyze: Genetic changes in tumors You work on: ?

Task 16. COLLABORATE. Work in small groups. You are a group of bioinformatics interns working in the genetics department of a hospital. While reviewing a routine genome scan, you discover something strange in a patient's DNA. It’s a short sequence in a non-coding region, where mutations usually don’t matter. But this sequence keeps appearing in different patients’ files. And it looks… unusual. When converted creatively into binary code, it seems to spell something. Some scientists say it’s just an error. Others believe it might be a hidden message or even part of a long-forgotten experiment. Discuss:

- Could it be a mutation?
- Could it be a software bug?
- Could someone or something have planted a message in the genome?



LANGUAGE FOCUS

Task 17. STUDY AND ANALYZE. Look at the rule about modals for probability, possibility and deduction, study in what situations they are used.

MODALS FOR PROBABILITY, POSSIBILITY AND DEDUCTION

USE	MODAL	EXAMPLE
Expressing certainty (or near certainty) about now or generally	must can't couldn't	That **must** be a genetically modified tomato. These **can't / couldn't** be natural plant genes; they were changed in a lab.
Expressing certainty (or near certainty) about the past	must can't couldn't (+ perfect infinitive)	Scientists **must have used** a computer to check the genetic changes in the modified animals. This program **can't / couldn't have made** a mistake in finding the modified genes.
Expressing probability about now, the future, or generally	should ought to	You **ought to / should** use software to predict how the modified animals will behave.
Expressing probability about the past	should ought to (+ perfect infinitive)	The tests **ought to / should have found** any problems before the crop was used.
Expressing possibility about now, the future, or generally	could may might	New software **could / may / might** help make genetically modified food healthier.
Expressing possibility about the real past	could may might (+ perfect	That **could / may / might have been** a mistake in editing the plant's DNA.

	infinitive)	
Expressing possibility about a hypothetical past	could might (+ perfect infinitive)	If researchers had used better technology, they **could / might have avoided** side effects in the modified animals.

Study these examples:

I'm sure he **knows** the data well. Perhaps the algorithm **will be** slow.	present infinitive	He **must know** the data well. The algorithm **may be** slow.
It's possible that he **is working** late tonight. I'm sure she **will be working** tomorrow.	present continuous infinitive	He **could be working** late tonight. She **must be working** tomorrow.
I'm sure he **didn't process** the data correctly. Perhaps they **have missed** the data. It's possible the technician **had made** an error.	perfect infinitive	He **can't have processed** the data correctly. They **might have missed** the data. The technician **may have made** an error.
I'm certain the software **was running.** Perhaps the algorithm **has been changing.** It's likely they **had been analyzing** the sequences.	perfect continuous infinitive	The software **must have been running.** The algorithm **may have been changing.** They could **have been analyzing the sequences.**

Task 18. PRACTICE. Complete the sentences using must or can't as in the example.
e.g. It's possible they skipped some data points during the experiment.
They **might have skipped** some data points during the experiment.

1. I'm sure they are working with a new database.
2. It's possible that the analysis was done yesterday.
3. Perhaps they haven't identified the mutation yet.
4. I'm certain she has finished her data analysis.
5. I'm sure they have already shared the results with the team.
6. It's possible the server went down last night.
7. I'm certain he checked the data before submitting it.
8. Perhaps they didn't consider all possible variables in the model.
9. It's likely they found an error in the sequence.
10. I'm sure they are using a faster processor for the simulation.

Task 19. PRACTICE. Rephrase the following sentences as many ways as possible.
e.g. It's likely that the researcher has forgotten to upload the data.
The researcher **may/might/could have forgotten** to upload data.

1. Perhaps the program will complete the task soon.
2. It's possible that they didn't use the correct algorithm.
3. It's likely they have not found the error in the code yet.
4. It's likely that the scientist has completed the sequencing analysis.

5. Perhaps the AI model will detect the mutation.
6. It's possible that the software crashed during the simulation.
7. It's likely that the scientist misinterpreted the genetic data.
8. Perhaps they use a different machine-learning model for the prediction.

Task 20. PRACTICE. Use the words in the box to complete the sentences in Table A. The meaning of the sentences in Table B will help you.

able ▪ cannot ▪ could ▪ have ▪ might ▪ must ▪ needn't ▪ mustn't ▪

Table A	Table B
1. I _________ have made an error in the DNA sequence analysis.	expressing certainty
2. In a few months, I'll be ___________ to use the new bioinformatics software.	expressing future ability
3. I __________ write simple scripts when I started working with biological data.	expressing past ability
4. No, you ___________ access the genetic database without permission.	refusing a request
5. You ________ forget to save your results before closing the software!	expressing personal obligation
6. You _________ analyze the entire genome—just focus on the key genes.	expressing a lack of obligation
7. You _________ to validate the data before publishing the research.	expressing future obligation
8. The algorithm __________ predict mutations in the genetic sequence.	expressing probability

Task 21. PRACTICE. Circle the correct word or phrase. If both are correct, circle both.

1. 'Did the system detect any errors?'
   '**That'll / That must** be due to a corrupted file.'
2. 'I spent all day annotating gene sequences.
   'You must **be / have been** exhausted. Take a break!'
3. 'The results should have been processed by now.'
   'It **mustn't / can't** be taking this long. Something is wrong.'
4. 'The software asked me to repeat every single data entry.'
   'You **must / can** have been frustrated.'
5. 'He submitted the report without checking the genetic model.'
   'He **can't / couldn't** have verified the data properly.'
6. 'I hope the new bioinformatics tool is available soon.'
   'They **may / might** still be testing it before release.'
7. 'The temperature settings in the lab are too high.'
   'Yes, I think it **must / might** affect the experiment.'
8. 'I can't wait to see the new research results!'
   'The findings **should / might** be published next week.'
9. 'I hope they find an efficient algorithm for genome analysis.'
   'We **should / might** be lucky and see a breakthrough soon.'
10. 'The database is not responding to my query.'
    'It **could / can't** be offline for this long – try restarting the system.'

UNIT 5.2 BIOINFORMATICS IN MEDICINE AND HEALTH

READING AND VOCABULARY

Task 22. READ FOR MAIN IDEA. Read the following paragraph titles for the article. Match them to the correct paragraph, one is not needed.

1. A New Era of Drug Development
2. Fighting Pandemics with Data
3. Coding for Cancer
4. Staying One Step Ahead
5. Data-Driven Healthcare
6. Personalized Treatment Plans
7. Genetic Clues to Rare Diseases

The Role of Bioinformatics in Medicine

A. Bioinformatics is transforming how doctors understand and treat diseases. Through the use of computer-based technology, scientists can now analyze a patient's genetic data to decide which treatments will work best. This new approach is called personalized medicine and is already helping patients with cancer, rare diseases, and even mental illnesses.

B. One of the biggest uses of bioinformatics in healthcare is detecting genetic disorders. By studying DNA sequences, doctors can identify gene mutations early, often before symptoms appear. This allows for early treatment and better results, especially in inherited diseases.

C. Another important area is drug discovery. It might take years to produce new medicine, but bioinformatics speeds it up. Computers can model the interaction between a new drug and the body, which saves time and money. It also makes it easier to design drugs against particular genes or proteins.

D. Bioinformatics is also changing how we fight infectious diseases. During a virus outbreak, scientists can analyze the genetic code of the virus rapidly in order to trace its spread and create vaccines. This technology was used during the COVID-19 pandemic to understand how the virus was changing.

E. In hospitals, bioinformatics tools are used to manage huge amounts of patient data. Electronic health records, test results, and medical histories may be reviewed in order to look for patterns and aid in decision-making. This helps doctors give better care and reduce medical errors.

F. In the future, bioinformatics has the potential to anticipate diseases before they actually happen. Provided with enough information, computers would be able to alert doctors in the future about possible health dangers based on a person's lifestyle and genetics, enabling preventive treatment and extended, healthier lifespans.

(adapted from https://www.genome.gov/genetics-glossary/Bioinformatics)

Task 23. READ FOR DETAILS. Read the following statements and decide if they are True, False or Not given.

1. Bioinformatics helps predict how drugs will interact with the human body.	True / False / Not Given
2. The use of bioinformatics in mental health treatment is still limited.	True / False / Not Given
3. DNA analysis can help detect diseases before any symptoms appear.	True / False / Not Given
4. Computers help doctors make faster and more accurate decisions.	True / False / Not Given
5. Bioinformatics was not used during the COVID-19 pandemic.	True / False / Not Given

6. Only people with rare diseases benefit from personalized medicine.	True / False / Not Given
7. Electronic health records are stored using bioinformatics software.	True / False / Not Given
8. Future tools may help warn people about health risks before they get sick.	True / False / Not Given
9. Scientists no longer need labs to study viruses thanks to bioinformatics.	True / False / Not Given

Task 24. COMMUNICATE. Work in pairs. Discuss the following questions.

1. Which application of bioinformatics in medicine do you find most useful? Why?
2. Do you think personalized medicine will become common in the future?
3. What are the possible risks of using computers to decide on treatments?

Task 25. WORK WITH WORDS. Match the words to their definitions.

1. **epigenetics**	a) an approach that uses artificial intelligence tools to identify illnesses based on data
2. **biomarker**	b) finding changes or errors in DNA that can cause diseases
3. **mutation detection**	c) a specific molecule, gene, or protein that scientists aim at when designing a treatment
4. **clinical trial**	d) a situation where a virus spreads quickly among a large number of people
5. **AI-based diagnosis**	e) disease that is passed from parents to their children through genes
6. **therapeutic target**	f) research focused on understanding and treating illnesses that affect only a small number of people
7. **rare disease research**	g) the study of how gene activity is regulated without changing the DNA sequence itself
8. **virus outbreak**	h) the search for new medicines and treatments through research and testing
9. **drug discovery**	i) a biological sign, such as a molecule in blood, that indicates a condition or disease
10. **inherited disease**	j) a carefully designed study involving human participants to test the safety and effectiveness of a medical treatment

Task 26. WORK WITH WORDS. Complete the text with the words from the box. One word is extra and shouldn't be used.

Fighting a Rare Disease

inherited disease ▪ mutation detection ▪ therapeutic target ▪ epigenetics ▪ AI-based diagnosis ▪ drug discovery ▪ rare disease research ▪ clinical trial ▪ biomarker ▪

When Emma started showing unusual symptoms at the age of five, doctors struggled to find the cause. After months of tests, a specialist suggested using 1) ______________ technology to search for hidden genetic changes. The results revealed a rare mutation, pushing Emma's case into a 2) ____________ program.

Researchers also investigated 3) _______________ to explain why Emma's symptoms were more severe than those of other children with the same mutation. They identified a 4) __________ – a molecule in her blood – that could help doctors diagnose similar cases more quickly in the future.

While Emma participated in a 5) ___________ for a new treatment, scientists worked on 6) ___________ projects to develop a permanent solution. In the meantime, an 7) ___________ system helped doctors track her condition and adjust her care plan. Thanks to these efforts, researchers are now focusing on a promising 8) ______________ that could one day lead to a full cure.

Task 27. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.
Crack the code – to solve a complex problem.
*The team **cracked the code** of the virus's genetic sequence.*
Needle in a haystack – something hard to find.
*Finding the mutation was like a **needle in a haystack**.*
Grow like weeds – to expand rapidly.
*Bioinformatics startups are **growing like weeds** this year.*
In your DNA – an inherent trait or skill.
*Analyzing data is **in her DNA** as a bioinformatician.*
Cut to the chase – to get to the point quickly.
*Let's **cut to the chase** and discuss the health application.*
Building blocks – fundamental components.
*Proteins are the **building blocks** bioinformatics relies on.*
Plant the seed – to start a process or idea.
*Her research **planted the seed** for a new medical tool.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. That early experiment _________ for a revolution in personalized medicine.
2. After months of analysis, they finally _________ and identified the disease's origin.
3. Searching for that rare protein in the dataset was like a _________, taking weeks to pinpoint.
4. His knack for solving bio-puzzles seems to be _________; he's a natural at it.
5. The professor said, "Enough theory – let's _________ and focus on curing this!"
6. DNA and RNA are the _________ that every bioinformatics model depends on.
7. New gene-editing companies __________ thanks to recent tech breakthroughs.

Task 28. COMMUNICATE. Tell the story of a bioinformatician's experiment that accidentally turns a lab rat into a super-smart mutant. Describe the surprising reveal. Use at least two idioms naturally.

Task 29. COLLABORATE. Work in pairs. Imagine a biotech lab where a data glitch stops a major gene project. How does the team recover? Use at least three idioms.

Task 30. WRITE. Write a short, funny story about two lab assistants failing a bioinformatics test, accidentally giving themselves mutant side-effects like super-speed. Use at least two idioms.

WATCHING

Task 31. WATCH FOR DETAILS. Scan the QR and watch the video "Genomic Medicine". Fill in the gaps in the sentences.

1. Genomic medicine uses the information in our __________ to inform and personalize our healthcare.
2. Our DNA contains around ____________ billion letters of code.
3. In the remaining ____________ % DNA, there are differences that can sometimes be harmful.
4. Clinical scientists, geneticists, _____________, and clinicians will study our DNA and compare it to many others.
5. They can figure out what treatment or ________________ will work best.
6. The NHS has provided ________________ gene testing for patients for decades.
7. We help the NHS analyze the whole genomes of patients with suspected ___________ and specific cancers.
8. Doctors could tell us what conditions we might be ____________ to.

Task 32. ANALYZE. Put the steps below in the correct logical order based on the process described in the video.

- Scientists and doctors compare DNA differences to those found in many other people.
- A patient's DNA is analyzed for harmful differences.
- The NHS offers targeted gene testing to patients.
- New technology allows analysis of the entire genome, not just specific genes.
- Treatment or prevention plans are developed based on the genetic information.
- Researchers use collected genome data to improve the understanding of diseases.
- Patients with suspected rare diseases and specific cancers are selected for whole genome analysis.
- Doctors may use genetic information to predict disease risks before symptoms appear.
- Genomics England aims to make genomic testing available to every patient.

LISTENING

Task 33. LISTEN FOR DETAILS. Scan the QR code and listen to the report of Moderna's spokesperson. Choose the correct answers to the questions.

1. What was the main focus of Moderna's research?
   A. Developing a COVID-19 vaccine using bioinformatics.
   B. Creating a new mRNA flu vaccine with bioinformatics.
   C. Studying cancer treatments through DNA analysis.
   D. Designing a mobile app to track flu symptoms.

2. How did Moderna use bioinformatics in their research?
   A. To analyze genetic data from 30,000 flu patients.
   B. To manufacture vaccines faster.
   C. To test side effects of traditional vaccines.
   D. To create a social media campaign about flu risks.

3. Which gene is linked to severe flu?
   A. BRCA1
   B. IFITM3
   C. APOE
   D. HER2

4. What result was seen in phase 3 trials of mRNA-1010?
   A. 60% fewer flu cases overall.

B. The vaccine cured cancer in 10% of patients.
C. 95% of participants had severe side effects.
D. Recovery time was 5 days shorter.

5. How long did Moderna take to develop the new flu vaccine?
A. months
B. 2 years
C. 1 year
D. 3 weeks

6. What do doctors use to check IFITM3 gene levels?
A. A urine test
B. An X-ray machine
C. A simple blood test
D. A smartphone app

7. What percentage of trial participants had mild side effects?
A. 40%
B. 70%
C. Over 95%
D. Less than 5%

8. Which organizations did Moderna partner with?
A. World Health Organization and UNICEF
B. National Institutes of Health and universities
C. Google and Apple
D. NASA and SpaceX

9. What are Moderna's future goals?
A. Building hospitals in developing countries.
B. Creating a flu vaccine for pets.
C. Inventing a cure for diabetes.
D. Fighting RSV and cancer with the same technology.

10. What does Moderna's 'mRNA Access' program do?
A. Tests vaccines on animals.
B. Sells vaccines at lower prices.
C. Shares bioinformatics data with scientists globally.
D. Teaches coding to medical students.

Task 34. WORK WITH WORDS. Fill in the gaps in the sentences with the words from the box. There are two extra words you do not need to use.

mutations ▪ mRNA vaccine ▪ genetic data ▪ bioinformatics ▪ partnerships ▪ side effects ▪ phase 3 trials ▪ recovery time ▪ genes ▪ immune system

1. Moderna used _________ to study flu patients' DNA and virus information.
2. The new _________ teaches the body to fight the flu virus using messenger RNA.
3. Scientists compared _________ from 30,000 people to find the IFITM3 gene link.
4. The _________ helps protect you when you get sick by fighting infections.
5. In _________, the mRNA-1010 vaccine showed 40% fewer severe flu cases.
6. Patients with the IFITM3 gene had a longer _________ when they got the flu.
7. Most people in the trial only experienced mild _________, like a sore arm.
8. Moderna's work with universities is an example of successful _________.

Task 35. WORK WITH WORDS. Find and correct the **one wrong word** in each sentence.

1. Bioinformatics helps scientists design vaccines by studying genetic dates.
2. Phase 3 trials test new drugs on animals.
3. Mutations in the jean IFITM3 can cause severe flu.
4. Personalized medicine uses the same treatment for everyone.
5. The imune system protects the body from viruses.
6. The new vaccine reduced healing time by 5 days.
7. Scientists analyzed genic data from 30,000 patients.
8. Clinical trails ensure the vaccine is safe for humans.

SPEAKING

Task 36. COLLABORATE. Work in groups of 3–5. Put the 'true' and 'false' headings on the table. Place the rest of the cards face down in a pile. Take turns: one student draws a card, reads it out loud, and the group discusses whether it's true or false. Decide together and place the card in the correct pile. Check the answers; 1 point is given for each correct guess. Group with the most points wins.

Written and Unwritten Rules for Scientists

Card 1 Scientists must share patient genetic data with everyone.	**Card 9** Scientists only need to report successful AI experiments.
Card 2 Publishing research openly helps medical progress.	**Card 10** During virus outbreaks, it's common to work with researchers from other countries.
Card 3 It's OK to use AI tools for diagnosis without checking by doctors.	**Card 11** Scientists are allowed already bring extinct animals, like mammoths, back to life.
Card 4 People must agree freely before joining clinical trials.	**Card 12** Editing human DNA to stop diseases is legal everywhere.
Card 5 Scientists can change published articles without telling the journal.	**Card 13** Scientists usually wait for peer review before announcing new discoveries.
Card 6 Researchers often share biological samples with trusted colleagues.	**Card 14** AI can now replace clinical trials to find rare diseases.
Card 7 If a scientist finds a new mutation, they should keep it secret to get more money.	**Card 15** Scientists should think about how their discoveries affect society.
Card 8 Making new species from different organisms is normal.	**Card 16** In emergencies, scientists can sometimes skip normal rules about data sharing.
TRUE	**FALSE**

Task 37. COLLABORATE. At first work individually. Sort the 10 cards from most to least important for your future career. Arrange them in a pyramid. Then work in pairs. Compare your pyramids. Discuss similarities and differences. Finally work as a class. Share the most important and least important factors.

Priority Pyramids

Working on cutting-edge medical research	Access to powerful AI tools and databases
Recognition in scientific journals and conferences	Opportunities to collaborate internationally
Flexible working hours (remote work possibility)	Competitive salary in biotech industry
Freedom to explore your own research ideas	Supportive team and mentorship
Continuous learning and professional development	Clear career path into data science, genomics, or pharma

Task 38. ANALYZE. Read the case study and discuss possible solutions to the problem.

A biotechnology lab in South Korea has been working with a gene therapy platform that combines artificial intelligence and bioinformatics to support personalized treatment for rare diseases. The system is designed to analyze each patient's full genome and pinpoint mutation patterns that could guide the design of customized gene-editing approaches. At first, the platform worked well, especially during the early testing phase, when the number of patients was still fairly low. But as enrollment increased, the system began to slow down. It struggled to keep up with the rising volume of genomic data, and the delays started to affect report generation. What had been quick and responsive in a research setting was now noticeably lagging under clinical pressure. The infrastructure behind the platform turned out to be part of the problem. It had originally been set up for research applications, not for handling clinical-scale data loads in real time. Doctors on the project began voicing concerns, noting that delayed reports could disrupt treatment schedules, particularly in cases where timing really matters. The company, in turn, is under growing pressure to find ways to improve system performance but without losing the reliability and accuracy that make the platform valuable in the first place.

What adjustments (technical or structural) could help the lab increase performance without compromising the integrity of its results?

LANGUAGE FOCUS

Task 39. STUDY AND ANALYZE. Look at the rule about modals for advice and criticism, study in what situations they are used.

MODALS FOR ADVICE AND CRITISISM

USE	MODAL	EXAMPLE
Asking for and giving advice	Should, shouldn't, ought to, oughtn't to, had better, hadn't better (only in questions), may/might as well (used when there is no better option)	You **should** use software to analyze genetic data. **Hadn't you better** check if the AI found the right mutation? We **may as well** run another test since the results are unclear.
Criticising past behaviour	should /shouldn't +perfect infinitive,	You **shouldn't have ignored** the software warning about the DNA error.

	ought to/ oughtn't to +perfect infinitive	
Expressing annoyance at past behaviour	could +perfect infinitive, might +perfect infinitive	You **could have saved** time by using a program instead of doing it by hand.
Criticising general behaviour	will	He **will** always forget to update the database after tests.
Criticising a specific example of someone's general behaviour	would	You **would** delete the data just when we needed it.

Task 40. PRACTICE. Match the problems with advice.

1. The DNA data has mistakes.	a) You should make the analysis faster.
2. The AI gave different answers for the same case.	b) You should check the system logs for problems.
3. The software is old and missing new tools.	c) You should compare the test with more data.
4. The patient's genetic test is unclear.	d) You ought to test the method before publishing the study.
5. The drug database is incomplete.	e) You might as well use another drug database.
6. The data analysis is too slow.	f) You shouldn't have forgotten to record changes.
7. The team forgot to write down changes in the data.	g) You should check for errors in the data.
8. The hospital's system suddenly stopped working.	h) You ought to fix the protein model settings.
9. The protein structure prediction is wrong.	i) You ought to train the AI with better examples.
10. The research didn't test if the method works well.	j) You had better update the software.

Task 41. PRACTICE. Rewrite each sentence using a different modal verb.
e.g. It is important to update the genetic database.
You **should** update the genetic database.
1. It was a mistake to delete the raw sequencing data.
2. It's a good idea to double-check AI-generated drug predictions.
3. There is no better option than using machine learning for genome analysis.
4. Ignoring the system error was a bad decision.
5. It's necessary to validate the AI model before using it for medical research.
6. His skipping data cleaning steps often leads to errors.
7. It was possible for the researchers to detect the mutation earlier.
8. As usual, she ran the sequencing program without checking the settings.
9. It was a bad decision not to encrypt patient records.
10. As always, he forgot to back up the research data before running the program.

Task 42. PRACTICE. Each sentence below has a mistake in the use of modals. Find and correct it.

1. Scientists should to test all new drugs carefully.
2. You must skip vaccine safety trials.
3. The AI model could has predicted dangerous side effects.
4. You had better to check the vaccine's storage conditions.
5. Researchers didn't ought to publish results without checking them.
6. The team shouldn't had ignored patient feedback.
7. The system might can analyze thousands of vaccine samples.
8. You would have tested the new drug on animals before human trials.
9. The researchers may as well to use AI for faster drug discovery.
10. You can have tested the program before using it in real cases.

UNIT 5.3 BIOTECH IN SOCIETY

READING AND VOCABULARY

Task 43. READ FOR MAIN IDEA. You are going to read an article called ‘Human Tissue Meets Robotics’ about recent advancements in creating biohybrid robots. Tick the things that you think might be goals or outcomes of creating biohybrid robots. Then, read the article and check which points are actually mentioned.

- improving prosthetics
- making robots that can entertain people
- developing medical robots that move naturally
- creating robots with self-healing abilities
- building robots for space missions
- raising ethical questions
- inventing robots that can replace doctors

Task 44. READ FOR DETAILS. Read the text and choose the correct answers to the questions below.

Human Tissue Meets Robotics

Recent years have seen scientists in Japan make major strides towards creating robots that will perform more like living beings. They are accomplishing this by combining human biological tissue with machinery to create what are called biohybrid robots – robots which can move, react, and even show emotion using real human cells.

In 2025, scientists from the University of Tokyo and Waseda University developed a robotic hand powered by lab-grown human muscles. The muscles were cultivated using real human cells and grouped into tiny bundles called "MuMuTa" – a shortened form of Multiple Muscle Tissue Actuators. When an electric current is transmitted, the muscles contract, similar to how they do within the human body. This allows the robot hand to perform simple movements such as grasping or pointing. Although the hand now has only three degrees of freedom, the hand worked day and night for almost six months and performs best at body temperature – about 37°C.

A year back, scientists at the University of Tokyo achieved another milestone when they created a robot face with live human skin grafted over it. This skin was not glued on like a sheet of paper – it was intentionally fastened with a method imitating how natural human skin is affixed to muscle and bones. As a result, the robot could make normal facial expressions, including smiling and frowning. The skin also exhibited the ability to self-heal when lightly injured, similar to natural skin.

These innovations represent a significant step in the emerging area of biotechnology, which merges biology and engineering. The aim is not to create robots more human in order to entertain, but to enable new opportunities in medicine, prosthetics, and rehabilitation. Consider artificial limbs that react similarly to biological ones or robots that are able to assist patients through natural gestures and facial expressions.

As biohybrid systems continue to advance, the line between living tissue and machine erodes – conjuring not only technological potential but also ethical questions about the future of human-robot interaction.

(adapted from https://mikekalil.com/blog/japan-human-muscle-robot-hand/)

1. What are biohybrid robots?
   A. Robots made of metal and biopolymers
   B. Robots using both human tissues and machines
   C. Robots controlled by biological organisms and AI
   D. Robots that live in natural environments
2. What powers the robotic hand developed in 2025?

    A. Solar panels
    B. Artificial muscles
    C. Lab-grown human muscles
    D. Mechanical springs
3. What does "MuMuTa" stand for?
    A. Muscle Mutation Technology Actuators
    B. Multiple Muscle Tissue Actuators
    C. Multi-Movement Training Algorithm
    D. Mechanical Muscle Tension Actuator
4. How does the robotic hand move?
    A. By solar energy
    B. By hydraulic pressure
    C. By wireless remote control
    D. By electric signals
5. What was developed in 2024?
    A. A robot leg with muscles
    B. A robot arm for surgery
    C. A robot face with human skin
    D. A robot brain for learning
6. How was the robot skin attached?
    A. Inspired by human body
    B. Using glue
    C. Using metal wires
    D. By freezing the skin onto the surface
7. What future applications are mentioned for biohybrid robots?
    A. Creating robot pets
    B. Improving prosthetics and rehabilitation
    C. Building robots for sports competitions
    D. Making human-like toys
8. What concern does the article raise about biohybrid robots?
    A. Their high cost
    B. The lack of human parts for robots
    C. Human-robot interaction
    D. The risk of human viruses in robots

Task 45. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Why do you think the scientists used real human cells in their robotic hand and face?
2. What medical applications could biohybrid robots have in the future?
3. How does the robotic hand mimic human movement? What makes it different from a mechanical hand?
4. Would you feel comfortable interacting with a robot that has living human tissue? Why or why not?
5. In your opinion, should there be limits on using real human tissue in robotics? Why?
6. How might these technologies change prosthetics and rehabilitation in the next 10 years?
7. Do you think robots with human-like expressions would be more helpful or more disturbing?

Task 46. WORK WITH WORDS. Match the two parts of collocations from the text above.

1. **lab-grown**	a) movements
2. **make**	b) human muscles

3. **grouped into**	c) facial expressions
4. **perform**	d) injured
5. **degrees**	e) of freedom
6. **achieve**	f) limbs
7. **normal**	g) major strides
8. **lightly**	h) interaction
9. **artificial**	i) a milestone
10. **human-robot**	j) tiny bundles

Task 47. WORK WITH WORDS. Read three short texts and fill in the words from the box. One word is extra and shouldn't be used.

social responsibility ▪ biodegradable ▪ GMOs ▪ labeling laws ▪ gene editing ▪ bioethics ▪ pharming ▪ species ▪ biomanufacturing ▪ public perception ▪

Fighting Vitamin Deficiency

Golden Rice is a special type of biotech crop developed to help people in countries where many suffer from vitamin A deficiency. Scientists used 1) ______________ to add genes from other plants, so the rice produces more vitamin A. This can prevent serious health problems like blindness. Even though Golden Rice can save lives, 2) ___________ is mixed. Some consumers want strong 3) ______________ because they worry about eating genetically modified foods without knowing it.

Editing Mosquito Genes

In Africa, scientists are using biotechnology to fight malaria. They have edited mosquito genes so the insects cannot carry the malaria parasite. This method could save millions of lives. However, ethical concerns are growing. Some people fear changing the environment permanently. 4) ________ experts are asking if humans should have the right to change whole 5) ___________. Genetic privacy is less of a problem here, but 6) ____________ – thinking about future generations – is very important.

Biomanufacturing

A biotech startup in the USA is using bacteria to create 7) ____________ materials. These materials can replace plastic, which often pollutes oceans. The company practices 8) ______________ by programming bacteria to grow materials that naturally break down. This supports sustainability and reduces waste. However, strict biotech regulations are needed to make sure these new materials are safe for health and the environment. 9) _______________ is also growing: some companies are making medicines inside plants instead of traditional factories.

Task 48. COMMUNICATE. Work with a partner. Answer the questions below.

1. Do you think genetically modified food should always be labeled? Why or why not?
2. Would you personally eat genetically modified foods like Golden Rice? Why or why not?
3. What are the possible risks of eating genetically modified foods?
4. Should humans be allowed to change the genes of wild species? Why or why not?
5. In your opinion, what is more important: saving lives now or protecting nature for the future?
6. How can scientists balance innovation with social responsibility?
7. Would you trust products made by bacteria? Why or why not?
8. Why are biotech regulations important when creating new materials?
9. Do you think biomanufacturing could replace traditional factories in the future?

10. What are the advantages and disadvantages of using plants to produce medicines (pharming)?

Task 49. COLLABORATE. Work in small groups. Read about groundbreaking biotech startups. Prepare a 1-minute investor pitch explaining:

- what their startup does
- why it matters
- why someone should invest

One student from each group becomes an "investor". Investors vote for the most convincing project after all pitches.

1. Retro Biosciences (San Francisco, CA)

Sam Altman-supported Retro Biosciences is at the forefront of the use of artificial intelligence to extend human lifespan. Working with OpenAI, they've developed GPT-4b Micro, a specialized AI model that significantly increases efficiency in stem cell research. The model has achieved cellular reprogramming 50 times more effectively than current methods, a significant advancement in longevity research.

2. Constructive Bio (Cambridge, UK)

Constructive Bio is a company that produces new materials out of tiny living cells. Instead of creating plastics and drugs in massive factories, they "program" the cells to produce these in a greener and cleaner method. This minimizes pollution and wastage. They are also partnering with other companies in order to produce some drugs more efficiently.

3. Colossal Biosciences (Dallas, TX)

Colossal Biosciences is driving de-extinction efforts to bring back extinct animals like the woolly mammoth and the Tasmanian tiger. In 2025, they reported the birth of three successfully gene-edited wolf pups with characteristics of the now-extinct dire wolf. The development included gene-editing for 14 genes in gray wolves to express favored dire wolf traits before cloning and implantation into surrogacy dogs. These technologies not only push the frontiers of conservation and synthetic biology but also open up new frontiers for restoration of biodiversity.

4. Beckley Psytech (Oxford, UK)

Beckley Psytech is developing new medicines for mental disorders with special natural substances called psychedelics. Their medicine is a nasal spray that might cure people who have severe depression or alcohol problems. The treatment is now being tested to see if it works. The company plans to offer the new medicines to more people in the future.

5. BIOCAD (St. Petersburg, Russia)

BIOCAD is a leading Russian biotechnology company specializing in R&D, manufacturing, and clinical trials. The company specializes in developing cancer and autoimmune disease treatments and has market expansion plans for the European region. BIOCAD has developed seven medications for the European market, including new and biosimilar medicines to treat melanoma, breast, stomach, kidney, and lung cancers, rheumatoid arthritis, psoriasis, and multiple sclerosis.

LISTENING

Task 50. LISTEN FOR MAIN IDEA. Scan the QR code and listen to Vanessa's pitch about a biotech start-up working to protect the environment. Which option best describes her pitch?

A. Vanessa explains how bioinformatics is used to map endangered species and monitor ecosystem changes.
B. Vanessa presents software that uses bioinformatics and engineered microbes to break down plastic waste in nature.
C. Vanessa describes a platform that uses genetic data to predict a person's risk of developing rare diseases.

Task 51. LISTEN FOR DETAILS. Listen to the audio again. Complete the table below using the words you hear.

Start-up Name	1. EcoBio_______.
Problem	2. Pollution in water caused by ________, plastic, and chemicals.
Solution	3. Genetically modified ________ that break down harmful substances.
Software Used	4. A tool called ________.
Created By	5. A team of ______________ and data scientists.
First Step in Process	6. Compare DNA ________ from bacteria.
Next Step	7. Create a digital ________ of a new bacterial strain.
Testing	8. The models are tested in the ________.
Planned Field Trials	9. In two polluted ________ in Central Europe.
Goal	10. A cleaner ________, powered by science, technology and care.

Task 52. COMMUNICATE. Work in pairs. Role play a situation below.

Student A: You are Vanessa, the startup founder. You've developed a project that uses engineered microbes and bioinformatics software to break down plastic in polluted environments.

Use these points in your answers:

- What your startup does
- How the software works and who created it
- How bioinformatics helps your mission
- Real-world results (e.g. polluted beach case)
- Your future plans or vision
- Why your startup matters for the planet

Student B: You are an investor at a biotech competition. You are considering whether to give funding to Vanessa's startup. Ask detailed questions to understand how the company works and if it's a good investment.

You can ask about:

- The technology and how it works
- The team and their qualifications
- The results achieved so far
- How the startup plans to scale
- Any risks or challenges
- The role of bioinformatics in the project

Ask at least 5 questions, take notes, and decide if you will invest.

SPEAKING

Task 53. COLLABORATE. Work in groups of 14. You are going to read a strange, futuristic story connected to biotech. The story is divided into 14 parts but is out of order.

- Each student gets one sentence (or two students can share if needed).
- Memorize your sentence.
- Stand up and organize yourselves in the correct order by speaking aloud.
- Once in order, sit down and dictate your sentence to your group to write down the full story together.

Strange Things in the World of Biotech

Soon, scientists discovered that the eco-trees were also absorbing mobile signals and GPS data.	Citizens began to worry: were the eco-trees spying on them?
New laws were passed to control the	In court, the company argued that the eco-trees

development of bioengineered plants.	were never designed to spy on people.
One investigation team found that a hacker group had secretly modified the eco-trees' programming.	The eco-trees were not natural plants; they were built with living cells and tiny robots.
Governments quickly set up investigation teams to study the strange trees.	After the discovery, the company behind eco-trees faced international lawsuits.
Many cities removed the eco-trees and replaced them with real, natural trees.	Some eco-trees even started showing strange behavior, like moving slightly toward strong Wi-Fi signals.
Today, scientists continue to create new biotech, but society watches much more carefully.	It all started in 2035, when a biotech company grew the first "eco-tree" that could clean air much faster than real trees.
At first, people were excited and planted eco-trees in cities all around the world.	The hackers wanted to collect personal data to sell to advertising companies.

Task 54. COLLABORATE. Divide into two teams to have a debate about bioengineered plants. Some people believe bioengineered plants can solve global problems like hunger and pollution. Others worry about health risks, environmental damage, and loss of natural biodiversity. What do you think? Are bioengineered plants a great innovation or a dangerous experiment?

Bioengineered plants: Innovation or Danger?

Innovation	Danger
1. Bioengineered plants can grow faster and survive in poor conditions. 2.	1. Modified plants might spread uncontrollably and harm natural ecosystems. 2.

Task 55. COLLABORATE. Work in small groups. Think about the future. What would you like to have thanks to biotechnology? Fill in short notes under these topics.

Ideal Biotech World

Food	Health	Environment	Daily Life	Other
e.g. apples that don't rot 1. 2.	e.g. quick cure for flu 1. 2.	e.g. trees that clean air 1. 2.	e.g. clothes that fix themselves 1. 2.	e.g. robots that help old people 1. 2.

In groups of 3–4, talk about the world today and compare it to your ideal future.
Use these phrases to help:

- Today, we don't have enough…
- In the future, I hope we will have…
- Now it's difficult to…, but in the future…

Then each student chooses one person (secretly) from the list:

- A parent with a sick child
- A farmer

- A university student
- An environmental activist
- A biotech entrepreneur
- An elderly person

Talk about the biotech future like you are that person.
Useful phrases:

- For me, the most important thing is…
- I am worried about…
- I would like to have…

The others guess who you are.

Task 56. COLLABORATE. Match the comments to people's stories. Explain what the commenter thinks about each biotech.

Comments:

a) I don't trust these patches. What if they send my health data to companies without asking me?
b) Growing furniture sounds amazing – no factories, no pollution!
c) I'm happy I can eat fresh food even during hot summers. Biotech farming saves lives!
d) Technology is fantastic when it gives older people back their freedom and hobbies.
e) It's creative and eco-friendly, but what if grown furniture isn't as durable as traditional materials?
f) This sounds so convenient, but I wonder if depending on a patch could make you less aware of your real body needs.
g) It's a brilliant idea, but I'm worried that such plants could spread and change traditional farming forever.
h) It's amazing that you can move freely again, but are you ever afraid the implant might wear out too soon?

1. Which lifestyle would you prefer and why?
2. Which person is taking the biggest risk with biotech?
3. Which technology do you think will happen first?

Task 57. COLLABORATE. Work in pairs. Take three random cards. Invent a new biotechnology product that includes all three features from your cards. Think about how it could help society (in healthcare, environment, agriculture, etc.). Present your product to the class or walk around and "sell" it to others. Vote or buy one product – you cannot vote for or buy your own product. The product with the most votes or sales wins!

What's Your Biotech Solution?

laboratory-tested	biodegradable packaging
AI-assisted diagnostics	durable in harsh climates
prevents genetic diseases	one-year warranty
easy to produce at scale	environmentally friendly
portable and compact	reduces carbon emissions
rapid results in under 1 hour	affordable for developing countries
based on stem cell technology	safe for all ages
cutting-edge genetic engineering	reusable materials

Task 58. ANALYZE. Read the case study and discuss possible solutions to the problem.

A Japanese startup came up with an eco-friendly soil sensor made from natural polymers and cellulose. These small devices are meant to track things like moisture, temperature, and pH and then break down on their own after about six months in the ground. That's what made them so appealing: they didn't need to be collected or replaced manually. At first, everything seemed to be going well. But a few months in, some farmers using drip irrigation started noticing problems. Their sensors were failing after just three months much earlier than expected. It wasn't clear why at first, but lab testing showed that certain conditions, like higher soil acidity and heat, were speeding up the breakdown process. In other words, the very thing that made the sensors appealing (the way they degrade naturally) was happening too fast in real-world farm settings.

What adjustments could help make the sensors more durable in harsher conditions while keeping them biodegradable?

WATCHING

Task 59. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

stem cells – special cells that can develop into different types of body cells.
incompatibility – a situation where things cannot work well together.
CRISPR – a technology that can edit parts of DNA to correct mutations.
cardiac arrest – a sudden stopping of the heart's function.
muscular dystrophy – a genetic disease causing muscle weakness and loss
fossil fuels – traditional energy sources like coal, oil, and gas that pollute the environment.
algal biofuels – renewable fuels made from algae, considered cleaner than other biofuels.
corn-based feedstocks – crops like corn used to produce early-generation biofuels
oil spill – a leak of oil into the environment, especially into the sea
equity – fairness and equal access to opportunities or technologies

1. A major _____________ near the coast caused severe damage to marine life.
2. Scientists are studying how _____________ could replace damaged tissues in the future.
3. After the accident, the patient suffered _____________ and needed immediate help.
4. Using _____________, researchers managed to correct a dangerous genetic mutation.

5. Moving away from ____________ is essential to reduce greenhouse gas emissions.
6. One big challenge in organ transplants is ____________ between donor and recipient tissues.
7. Researchers are excited about ____________ because it offers a cleaner alternative to older biofuels.
8. ____________ is a serious condition that leads to progressive muscle weakening.
9. Early biofuels made from ____________ raised concerns about food supply and environmental impact.
10. Without careful planning, technological advances could increase the gap in ____________.

Task 60. WATCH FOR MAIN IDEA. Scan the QR code and watch the video "Biotech Breakthroughs". Match the biotechnological breakthrough to its field.

Environmental protection ▪ Healthcare ▪ Renewable energy ▪ Neurotechnology ▪

1. Regenerative medicine → ___________
2. Bioremediation → ___________
3. Gene editing → ___________
4. Personalized medicine → ___________
5. Algal biofuels → ___________
6. Brain-Computer Interfaces → ___________

Task 61. WATCH FOR DETAILS. Watch the video again and choose the correct variant to the questions.

1. What is a primary goal of regenerative medicine?
   A. Repairing or replacing damaged tissues and organs
   B. Developing new surgical tools for doctors
   C. Creating vaccines for infectious diseases
   D. Designing 3D-printed medicines
2. Which institution is mentioned for progress in lab-grown organs?
   A. Neuralink
   B. US Department of Energy
   C. Salk Institute
   D. New England Journal of Medicine
3. Personalized medicine is tailored using what?
   A. A patient's age and gender
   B. A patient's genetic profile
   C. A patient's dietary habits
   D. A patient's geographic location
4. What technology is Neuralink developing?
   Algal biofuels
   Lab-grown organs
   Brain-computer interface
   CRISPR gene-editing tools
5. Which generation of biofuels uses algae?
   A. First
   B. Second
   C. Third
   D. Fourth
6. What is bioremediation?

   A. Using bacteria to produce vaccines
   B. Editing genes to cure diseases
   C. Growing crops for renewable energy
   D. Cleaning pollutants with living organisms
7. What funding did the US Department of Energy allocate to algal biofuels?
   A. 10 million
   B. 15 million
   C. 18.8million
   D. 25 million
8. What can engineered bacteria do in bioremediation?
   A. Break down plastic waste
   B. Produce insulin for diabetics
   C. Enhance cognitive functions
   D. Replace damaged organs

WRITING

Task 62. WRITE. Write a formal letter of inquiry (100–120 words) to a biotech firm, medical research center, or health-tech startup. Imagine you are a university student specializing in IT, biology, or applied informatics. You are interested in gaining hands-on experience in bioinformatics.

Invent a company name and write about:

- Why you are writing (inquire about internship, research project, or lab placement)
- What area interests you (e.g., genome sequencing, personalized medicine, microbiome studies, AI in diagnostics)
- Your current studies and relevant skills (e.g., Python, databases, data visualization)
- Questions you want to ask (availability, how to apply, selection criteria)

Useful phrases:

- I am writing to inquire about…
- I am currently studying…
- I am especially interested in…
- Could you please provide information about…
- I would appreciate any additional advice on…
- I look forward to your reply.

LANGUAGE FOCUS

Task 63. STUDY AND ANALYZE. Look at the table about modals you have studied in previous units.

MODALS REVISION

USE	MODAL
Modals of ability	can, could, could +perfect infinitive, able to,
Modals for permission	can, can't , could, couldn't, may, might, mustn't, may not, wasn't allowed,
Modals for requests and offers	will, would you mind, shall, can, would you like me
Modals for obligation, necessity and prohibition	must, mustn't, have (got) to, need (to), don't have to, haven't got to, needn't, don't need (to), will have to, didn't have to, didn't need (to), needn't +perfect infinitive, can't
Modals for possibility, probability and deduction	must, can't, couldn't, couldn't + perfect infinitive, should ought to, should/outght to+ perfect infinitive, could, may, might, could/may/might+ perfect infinitive,

Modals for advice and criticism	Should, shouldn't, ought to, oughtn't to, had better, hadn't better (only in questions), may/might as well (used when there is no better option), should /shouldn't +perfect infinitive, ought to/ oughtn't to +perfect infinitive, could +perfect infinitive, might +perfect infinitive, will, would

Task 64. PRACTICE. Complete the sentences using the appropriate modal verb.

1. You __________ use DNA profiling tools in forensic investigations to identify suspects. (advice)
2. The database __________ help track genetic information about endangered species in the ecosystem. (probability)
3. The forensic team __________ cross-checked the genetic samples to avoid errors. (criticism for past behavior)
4. This mistake __________ avoided if proper DNA analysis techniques had been used. (possibility in the past)
5. The new tool __________ improve the detection of invasive species in ecosystems. (future probability)
6. Scientists __________ be more cautious when interpreting the genetic evidence in criminal cases. (general advice)
7. The researchers __________ used a better method to analyze the DNA sequences from the crime scene. (criticism of past behavior)
8. You __________ analyze the genetic data to confirm the species in the environmental samples. (ability now)
9. Conservationists __________ monitor biodiversity regularly to protect endangered species. (obligation)
10. If they had collected more samples, they __________ gathered more genetic data on the species. (hypothetical past ability)

Task 65. PRACTICE. Choose the correct variant.

1. ______ you please send me the latest genetic data report?
   A. Would
   B. Should
   C. Could
   D. Might
2. You __________ use DNA profiling tools in forensic investigations to identify suspects.
   A. must
   B. should
   C. will
   D. can't
3. Conservationists _______ manually update biodiversity databases if an automated system is available.
   A. a) don't have to
   B. b) must
   C. c) have to
   D. d) should
4. ______ we collaborate on the project that studies the impact of human activities on ecosystems?
   A. Should
   B. Will
   C. Shall

D. Must
5. ______ you mind checking the gene sequencing results for possible errors?
A. Would
B. Should
C. Must
D. Might
6. The lab _______ operate the new genome sequencing machine until all staff members are fully trained on it.
A. mustn't
B. don't have to
C. might
D. can
7. The technician ______ have made an error during the DNA extraction process, given the unexpected results.
A. must
B. would
C. could
D. should
8. If the research team had more funding, they ______ analyze the data faster and with more precision.
A. must
B. could
C. should
D. shouldn't
9. You ______ verify the DNA sample if the testing conditions remain the same as before.
A. don't have to
B. need to
C. must
D. should
10. Scientists __________ to be more cautious when interpreting the genetic evidence in criminal cases.
A. should
B. ought
C. will
D. can

Task 66. PRACTICE. Complete the text with the correct modal verbs. More than one variant is possible.

In 2019, police used bioinformatics to solve a crime. DNA was found at the crime scene, and scientists sequenced it. The analysis showed a partial match in a database. The match 1) ______ belong to a suspect already known to the police, but it wasn't enough to confirm the identity.

To improve accuracy, the scientists suggested they 2) ______ run more tests, including comparing the DNA with family members. The bioinformatics software 3) ______ quickly analyze large datasets and provide better results. They realized they 4) ______ cross-reference the sample with other database entries to ensure accuracy, but redoing the entire analysis was unnecessary.

Later, a more complete DNA sample was found, confirming the suspect's identity. The suspect 5) ______ have left no trace, but the DNA led to their arrest. The forensic team knew that they 6) ______ review the results carefully before making a final decision, but they were confident about the outcome.